



Analog IC Design

CMOS Technology

MOHAMMED NASR EL-DIN

Section 3 - Lab 08

Negative Feedback

Closed Loop OP Parameters

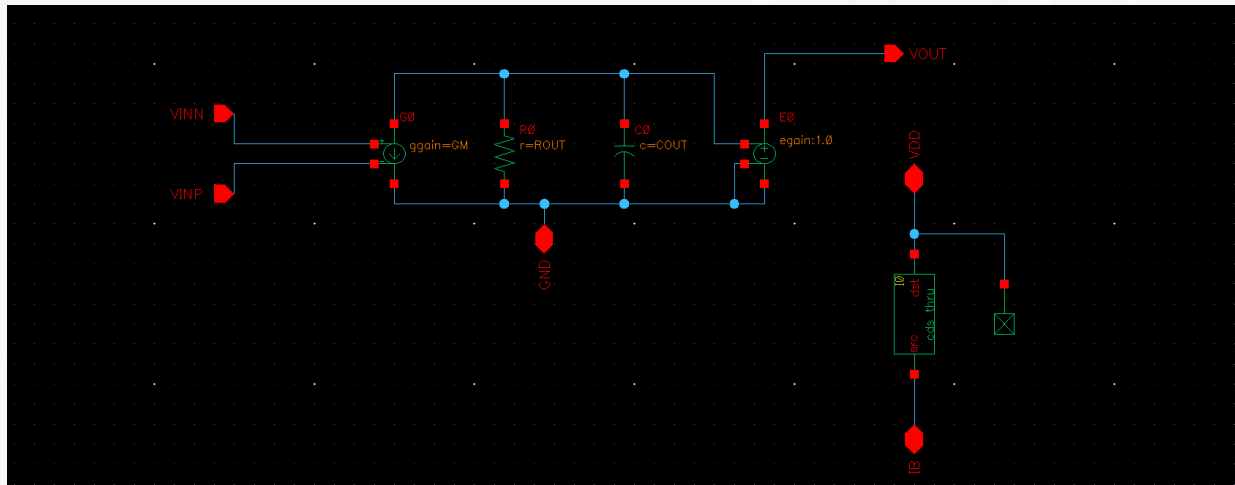
lab8:5T_OTA_tb:1	M1 ID	12.62u			
lab8:5T_OTA_tb:1	M1 vgs	925.9m			
lab8:5T_OTA_tb:1	M1 vds	389.1m			
lab8:5T_OTA_tb:1	M1 vth	845.7m			
lab8:5T_OTA_tb:1	M1 vdsat	135.7m			
lab8:5T_OTA_tb:1	M1 gm	160.4u			
lab8:5T_OTA_tb:1	M1 gds	1.773u			
lab8:5T_OTA_tb:1	M1 gmb	46.25u			
lab8:5T_OTA_tb:1	M1 gm/Id	12.71			
lab8:5T_OTA_tb:1	M1 vstar	157.4m			
lab8:5T_OTA_tb:1	M1 region	2			
lab8:5T_OTA_tb:1	M2 ID	12.62u			
lab8:5T_OTA_tb:1	M2 vgs	925.9m			
lab8:5T_OTA_tb:1	M2 vds	389.1m			
lab8:5T_OTA_tb:1	M2 vth	845.7m			
lab8:5T_OTA_tb:1	M2 vdsat	135.7m			
lab8:5T_OTA_tb:1	M2 gm	160.4u			
lab8:5T_OTA_tb:1	M2 gds	1.773u			
lab8:5T_OTA_tb:1	M2 gmb	46.25u			
lab8:5T_OTA_tb:1	M2 gm/Id	12.71			
lab8:5T_OTA_tb:1	M2 vstar	157.4m			
lab8:5T_OTA_tb:1	M2 region	2			

lab8:5T_OTA_tb:1	M3 ID	-12.62u			
lab8:5T_OTA_tb:1	M3 vgs	-986.9m			
lab8:5T_OTA_tb:1	M3 vds	-986.9m			
lab8:5T_OTA_tb:1	M3 vth	-763.5m			
lab8:5T_OTA_tb:1	M3 vdsat	-234.2m			
lab8:5T_OTA_tb:1	M3 gm	94.92u			
lab8:5T_OTA_tb:1	M3 gds	1.315u			
lab8:5T_OTA_tb:1	M3 gmb	37u			
lab8:5T_OTA_tb:1	M3 gm/Id	-7.519			
lab8:5T_OTA_tb:1	M3 vstar	-266m			
lab8:5T_OTA_tb:1	M3 region	2			
lab8:5T_OTA_tb:1	M4 ID	-12.62u			
lab8:5T_OTA_tb:1	M4 vgs	-986.9m			
lab8:5T_OTA_tb:1	M4 vds	-986.9m			
lab8:5T_OTA_tb:1	M4 vth	-763.5m			
lab8:5T_OTA_tb:1	M4 vdsat	-234.2m			
lab8:5T_OTA_tb:1	M4 gm	94.92u			
lab8:5T_OTA_tb:1	M4 gds	1.315u			
lab8:5T_OTA_tb:1	M4 gmb	37u			
lab8:5T_OTA_tb:1	M4 gm/Id	-7.519			
lab8:5T_OTA_tb:1	M4 vstar	-266m			
lab8:5T_OTA_tb:1	M4 region	2			

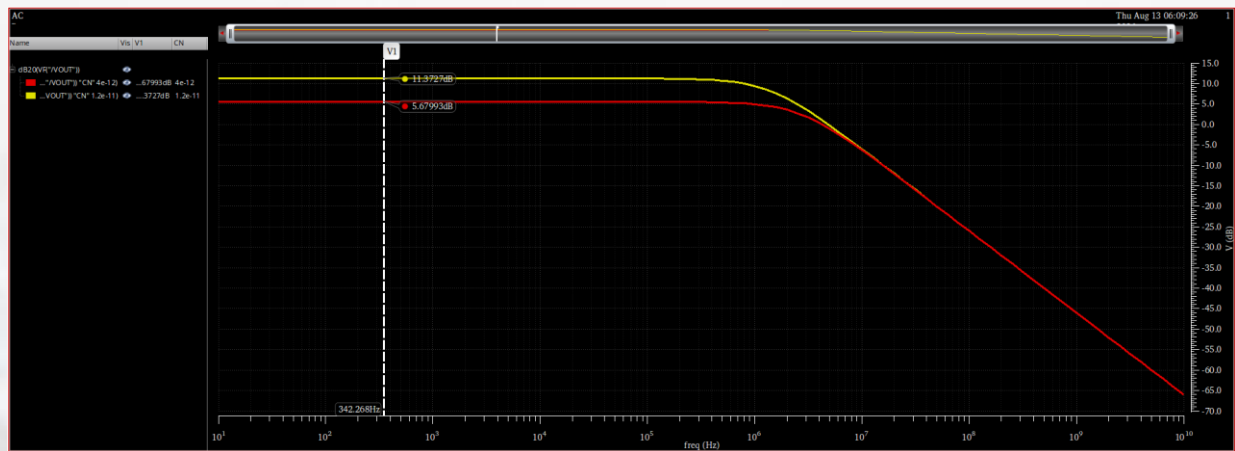
lab8:5T_OTA_tb:1	M5 ID	25.25u			
lab8:5T_OTA_tb:1	M5 vgs	930.1m			
lab8:5T_OTA_tb:1	M5 vds	424.1m			
lab8:5T_OTA_tb:1	M5 vth	698.7m			
lab8:5T_OTA_tb:1	M5 vdsat	238.1m			
lab8:5T_OTA_tb:1	M5 gm	177.5u			
lab8:5T_OTA_tb:1	M5 gds	2.532u			
lab8:5T_OTA_tb:1	M5 gmb	65.72u			
lab8:5T_OTA_tb:1	M5 gm/Id	7.032			
lab8:5T_OTA_tb:1	M5 vstar	284.4m			
lab8:5T_OTA_tb:1	M5 region	2			
lab8:5T_OTA_tb:1	M6 ID	10u			
lab8:5T_OTA_tb:1	M6 vgs	930.1m			
lab8:5T_OTA_tb:1	M6 vds	930.1m			
lab8:5T_OTA_tb:1	M6 vth	692.4m			
lab8:5T_OTA_tb:1	M6 vdsat	241m			
lab8:5T_OTA_tb:1	M6 gm	69.61u			
lab8:5T_OTA_tb:1	M6 gds	434.4n			
lab8:5T_OTA_tb:1	M6 gmb	25.78u			
lab8:5T_OTA_tb:1	M6 gm/Id	6.961			
lab8:5T_OTA_tb:1	M6 vstar	287.3m			
lab8:5T_OTA_tb:1	M6 region	2			

Same as in Lab7

Feedback With Behavioral OTA



Closed loop gain vs frequency



$V_{out}(dB)$ ■ $C_{IN} = 4pF$ ■ $C_{IN} = 12pF$

Point	Test	Output	Nominal	Spec	Weight	Pass/Fail
Parameters: CN=4p						
1	lab8:5T_OTA_ckt:1	VOUT				
1	lab8:5T_OTA_ckt:1	gain dB	5.68			
1	lab8:5T_OTA_ckt:1	gain	1.923			
1	lab8:5T_OTA_ckt:1	BW	2.649M			
1	lab8:5T_OTA_ckt:1	GBW	5.107M			
Parameters: CN=12p						
2	lab8:5T_OTA_ckt:1	VOUT				
2	lab8:5T_OTA_ckt:1	gain dB	11.37			
2	lab8:5T_OTA_ckt:1	gain	3.704			
2	lab8:5T_OTA_ckt:1	BW	1.376M			
2	lab8:5T_OTA_ckt:1	GBW	5.11M			

Compare the DC gain, BW, and GBW with hand analysis in a table

$$A_{ol} = 50, BW_{ol} = \frac{5GHz}{50} = 100KHz. A_{cl} \approx \frac{1}{\beta} = 1 + \frac{C_{IN}}{C_F}, BW_{cl} = (1 + A_{ol}\beta) BW_{ol}, GBW = A_{cl}BW_{cl}$$

For $C_{IN} = 4pF$:

$$A_{cl} = 1 + \frac{4pF}{4pF} = 2 = 6dB, BW_{cl} = \left(1 + 50 \times \frac{1}{2}\right) \times 100KHz = 2.6MHz, GBW = 2 \times 2.6MHz = 5.2MHz$$

For $C_{IN} = 12pF$:

$$A_{cl} = 1 + \frac{12pF}{4pF} = 4 = 12dB, BW_{cl} = \left(1 + 50 \times \frac{1}{4}\right) \times 100KHz = 1.35MHz, GBW = 4 \times 1.35MHz = 5.4MHz$$

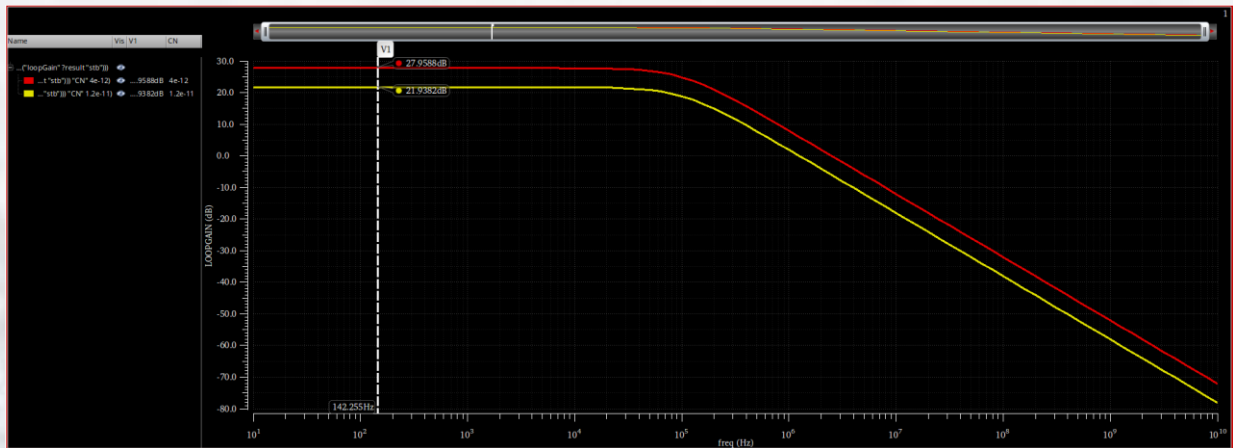
Parameter	$C_{IN} = 4pF$		$C_{IN} = 12pF$	
	Simulation	Hand calculations	Simulation	Hand calculations
DC gain	5.68dB	6dB	11.37dB	12dB
BW	2.65MHz	2.6MHz	1.37MHz	1.35MHz
GBW	5.11MHz	5.2MHz	5.11MHz	5.4MHz

Comment on the difference between the results for the two values of C_{IN}

As C_{IN} increase $\Rightarrow \beta \downarrow \Rightarrow (A_{cl} \propto \frac{1}{\beta}) \uparrow, (BW_{cl} \propto \beta) \downarrow \Rightarrow GBW = A_{cl} \times BW_{cl} = \text{constant with } \beta \text{ "C}_{IN}"$

For $C_{IN} = 12 \Rightarrow A_{cl}$ higher than in case $C_{IN} = 4$, and BW_{cl} smaller than in case $C_{IN} = 4$, but both have the same GBW.

Loop gain vs frequency



LoopGain (dB) ■ $C_{IN} = 4pF$ ■ $C_{IN} = 12pF$

Point	Test	Output	Nominal	Spec	Weight	Pass/Fail
Parameters: CN=4p						
1	lab8:5T_OTA_ckt:1	UGF	2.552M			
1	lab8:5T_OTA_ckt:1	LG	25			
1	lab8:5T_OTA_ckt:1	LG dB	27.96			
Parameters: CN=12p						
2	lab8:5T_OTA_ckt:1	UGF	1.272M			
2	lab8:5T_OTA_ckt:1	LG	12.5			
2	lab8:5T_OTA_ckt:1	LG dB	21.94			

Compare DC LG and GBW with hand analysis in a table

$$A_{ol} = 50, BW_{ol} = \frac{5\text{GHz}}{50} = 100\text{KHz}. LG = \beta A_{ol}, GBW_{LG} = LG \times BW_{ol}$$

For $C_{IN} = 4\text{pF}$:

$$LG = \frac{1}{2} \times 50 = 25 \approx 27.96\text{dB}, GBW_{LG} = 25 \times 100\text{KHz} = 2.5\text{MHz}$$

For $C_{IN} = 12\text{pF}$:

$$LG = \frac{1}{4} \times 50 = 12.5 \approx 21.94\text{dB}, GBW_{LG} = 12.5 \times 100\text{KHz} = 1.25\text{MHz}$$

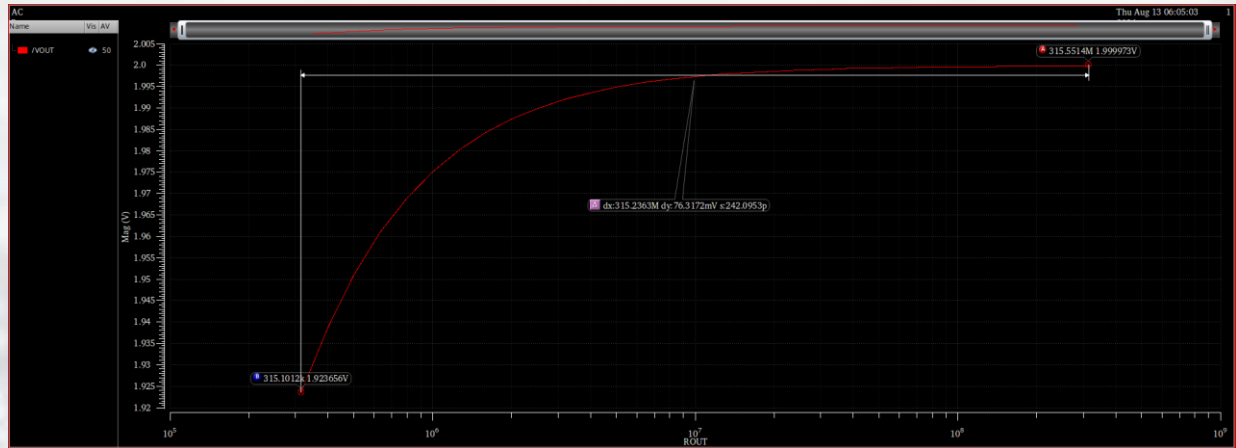
Parameter	$C_{IN} = 4\text{pF}$		$C_{IN} = 12\text{pF}$	
	Simulation	Hand calculations	Simulation	Hand calculations
DC LG	27.96dB	27.96dB	21.94dB	21.94dB
GBW	2.55MHz	2.5MHz	1.27MHz	1.25MHz

Comment on the differences between the results for the two values of C_{IN} .

As C_{IN} increase $\Rightarrow \beta \downarrow \Rightarrow (LG \propto \beta) \downarrow, (GBW_{LG} \propto LG) \downarrow$

For $C_{IN} = 4 \Rightarrow LG$ higher than in case $C_{IN} = 12$, and GBW_{LG} larger than in case $C_{IN} = 12$, but both have the same BW.

Gain Desensitization



A_{vcl} vs A_{vol}

Calculate the percent change in closed loop gain

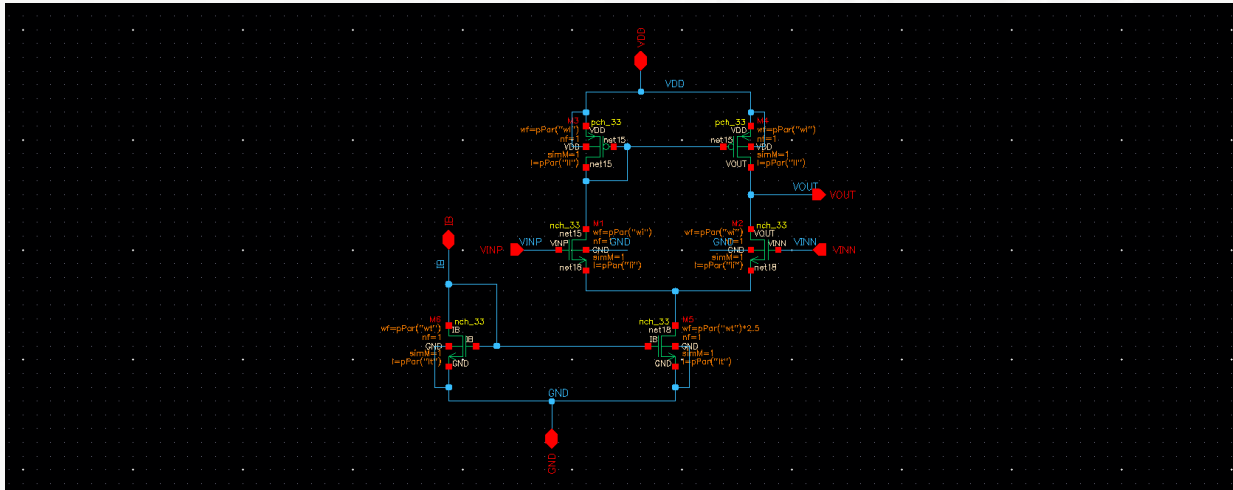
$$\%change = \frac{A_{vcl-max} - A_{vcl-min}}{A_{vcl-min}} \times 100\% = \frac{0.076}{1.92} \times 100\% = 3.96\%$$

Comment

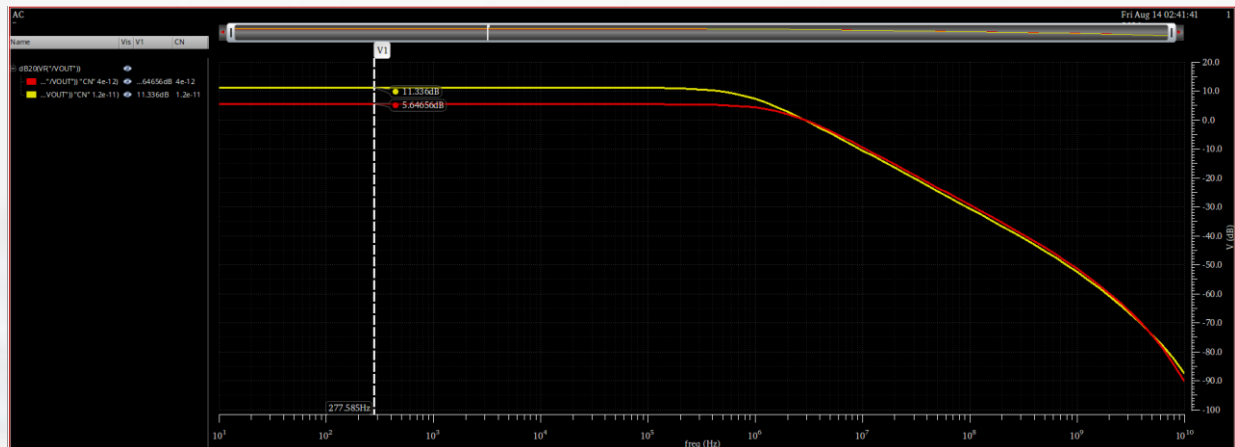
For large variation in open loop gain there is **small** variation in closed loop gain due to feedback effect “**gain desensitization**”

Feedback With Real OTA

Note: DC OP reviewed from result browser and it is same as in lab 7



Closed loop gain vs frequency


$$V_{out}(dB) \quad \blacksquare C_{IN} = 4pF \quad \blacksquare C_{IN} = 12pF$$

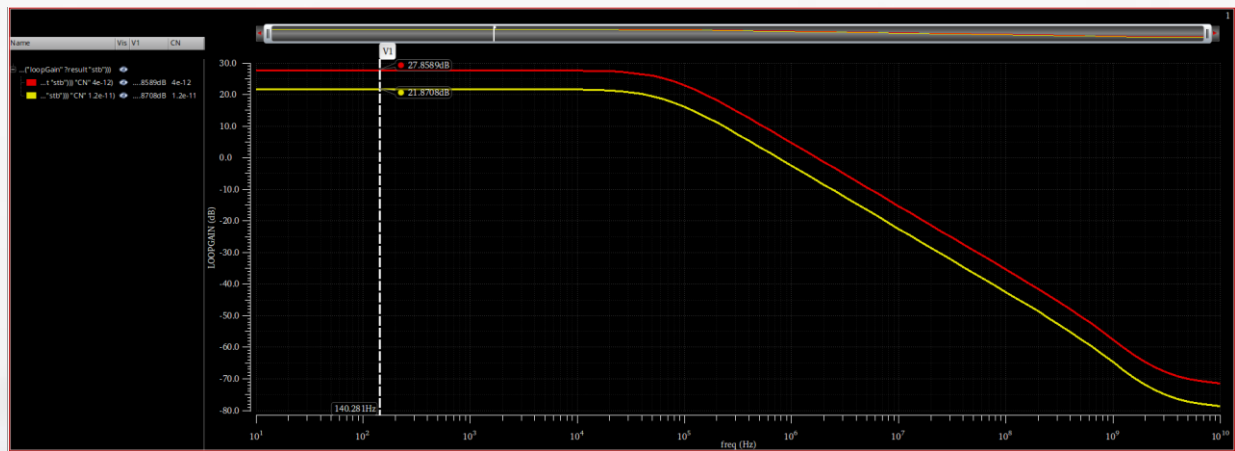
Point	Test	Output	Nominal	Spec	Weight	Pass/Fail
Parameters: CN=4p						
1	lab8:5T_OTA_ckt:1	VOUT				
1	lab8:5T_OTA_ckt:1	gain dB	5.647			
1	lab8:5T_OTA_ckt:1	gain	1.916			
1	lab8:5T_OTA_ckt:1	BW	1.821M			
1	lab8:5T_OTA_ckt:1	GBW	3.496M			
Parameters: CN=12p						
2	lab8:5T_OTA_ckt:1	VOUT				
2	lab8:5T_OTA_ckt:1	gain dB	11.34			
2	lab8:5T_OTA_ckt:1	gain	3.688			
2	lab8:5T_OTA_ckt:1	BW	826.6k			
2	lab8:5T_OTA_ckt:1	GBW	3.056M			

Parameter	Behavioral OTA		Real OTA	
	$C_{IN} = 4pF$	$C_{IN} = 12pF$	$C_{IN} = 4pF$	$C_{IN} = 12pF$
DC gain	5.68dB	11.37dB	5.65dB	11.34dB
BW	2.65MHz	1.37MHz	1.82MHz	0.85MHz
GBW	5.11MHz	5.11MHz	3.5MHz	3.1MHz

You will notice that the bandwidth, and consequently the GBW are much smaller than Part 1 Why? Comment

Since the BW is limited by the dominant pole at the high impedance node “output node”, there is a parasitic capacitance C_P that we didn’t model it for the behavioral model that its effect appears as the frequency getting higher, and in transistor level $BW \propto GBW \propto \frac{1}{C_{out}+C_P}$, so as frequency increases BW and GBW decreases.

Loop gain vs frequency



LoopGain (dB) ■ $C_{IN} = 4pF$ ■ $C_{IN} = 12pF$

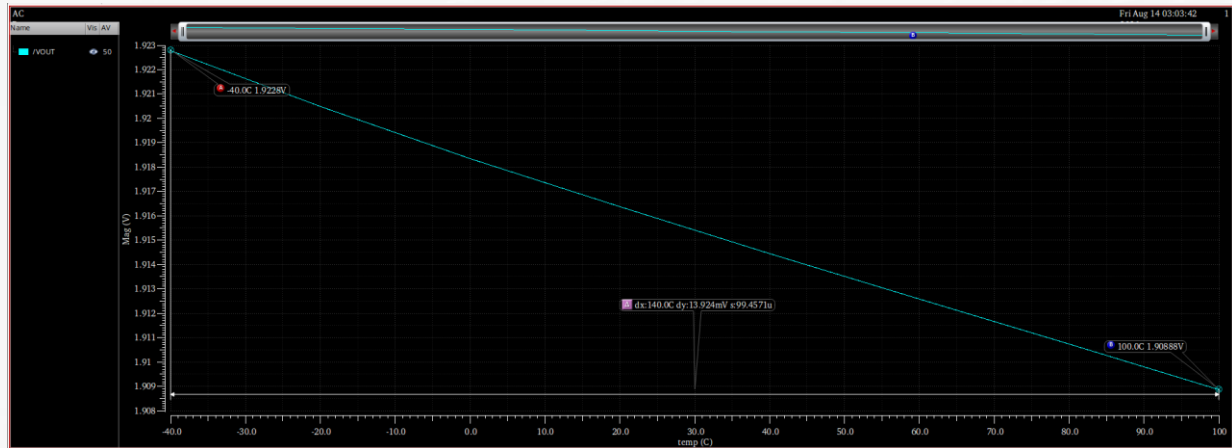
Point	Test	Output	Nominal	Spec	Weight	Pass/Fail
Parameters: CN=4p						
1	lab8:5T_OTA_ckt:1	UGF	1.768M			
1	lab8:5T_OTA_ckt:1	LG	24.71			
1	lab8:5T_OTA_ckt:1	LG dB	27.86			
Parameters: CN=12p						
2	lab8:5T_OTA_ckt:1	UGF	767.6k			
2	lab8:5T_OTA_ckt:1	LG	12.4			
2	lab8:5T_OTA_ckt:1	LG dB	21.87			

Parameter	Behavioral OTA		Real OTA	
	$C_{IN} = 4pF$	$C_{IN} = 12pF$	$C_{IN} = 4pF$	$C_{IN} = 12pF$
DC LG	27.96dB	21.94dB	27.86dB	21.87dB
UGF	2.55MHz	1.27MHz	1.77MHz	0.77MHz

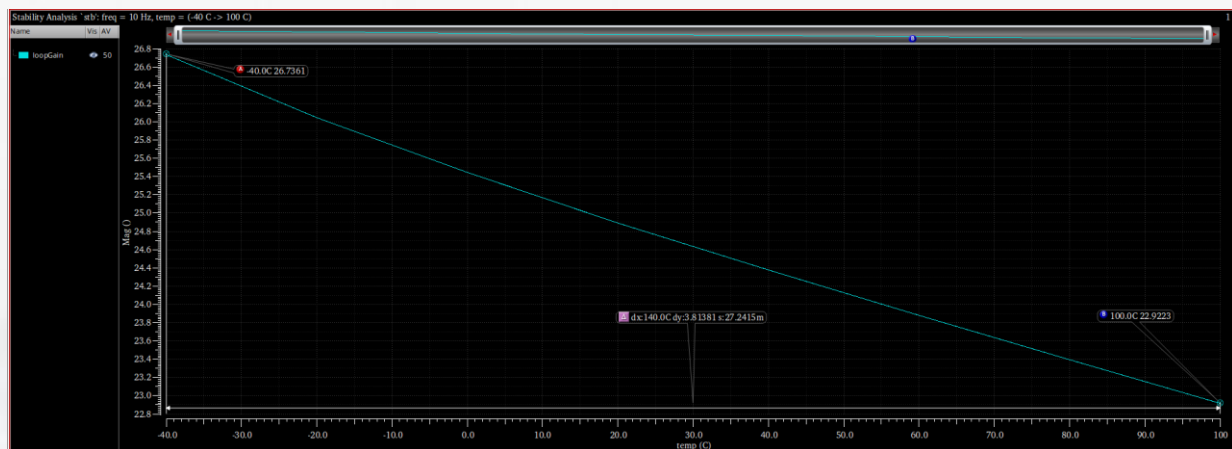
You will notice that the unity gain frequency is much smaller than Part 1 Why? Comment

Since the UGF limited by the first non-dominant pole that we didn’t model it in the behavioral model “we only modeled the dominant pole in the behavioral model”, the first non-dominant pole decreases for the same reason as the dominant pole “parasitic capacitance C_P ” and the UGF so.

Gain Desensitization



A_{vcl} vs temperature



LG vs temperature

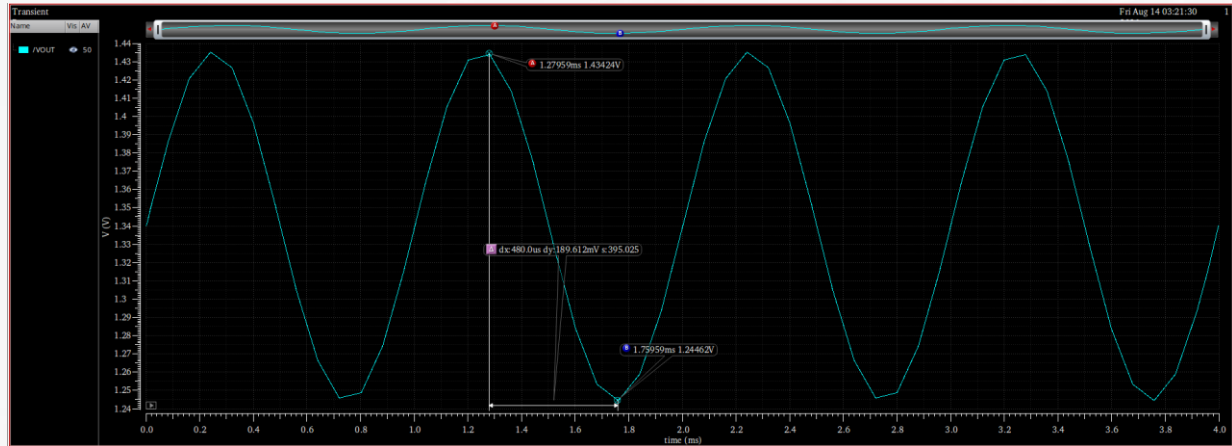
Compare the percent change in the DC loop gain and the DC closed loop gain across temperature extremes, comment

$$\text{Closed loop gain \%change} = \frac{A_{vcl-max} - A_{vcl-min}}{A_{vcl-min}} \times 100\% = \frac{13.9 \times 10^{-3}}{1.92} \times 100\% = 0.73\%$$

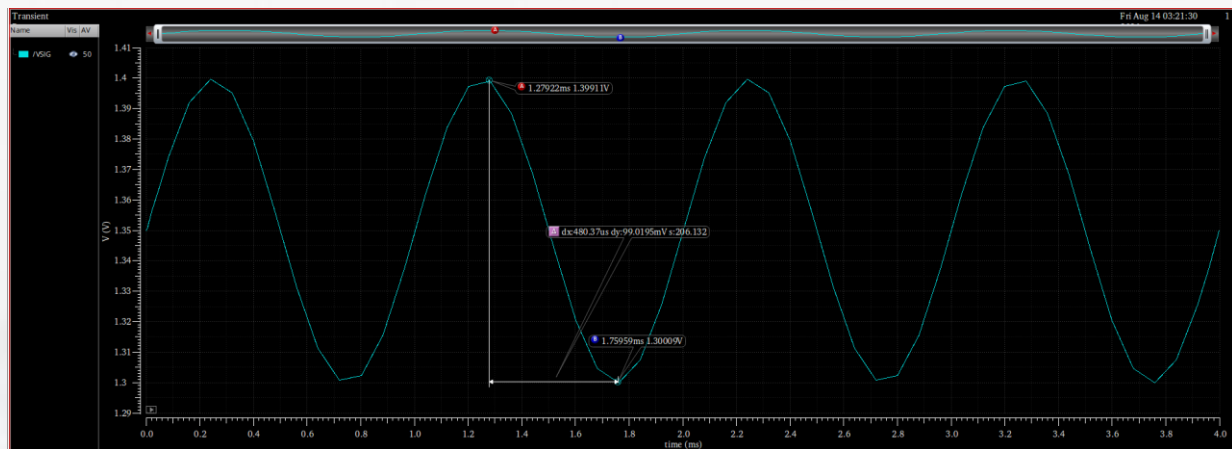
$$\text{Loop gain \%change} = \frac{LG_{max} - LG_{min}}{LG_{min}} \times 100\% = \frac{3.81}{22.92} \times 100\% = 16.62\%$$

The percent change in LG with the temperature is very large relative to the closed loop gain percent change, and this is a powerful advantage of negative feedback

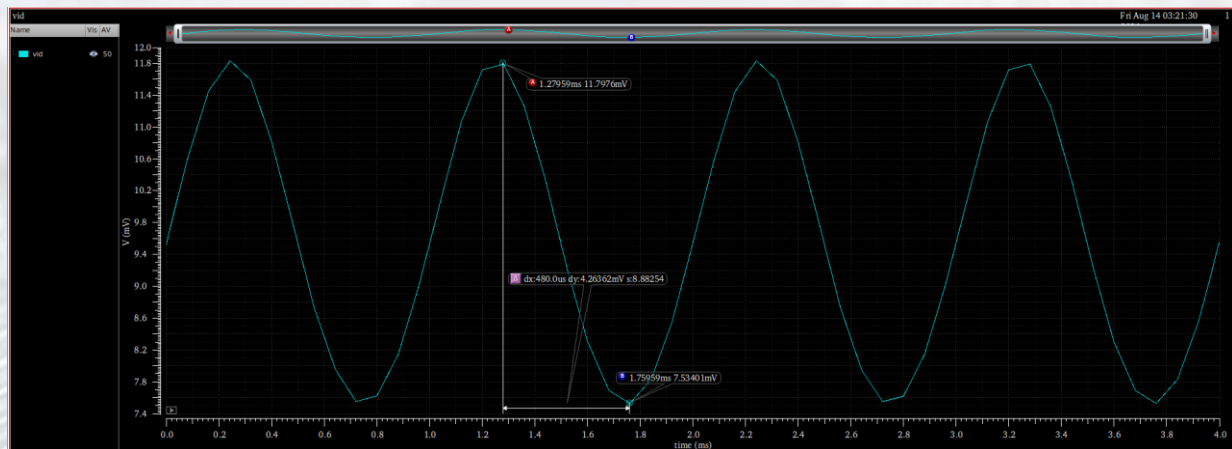
Transient analysis



V_{out} : P_2P voltage $\approx 190mv$



V_{sig} : P_2P voltage $\approx 100mv$

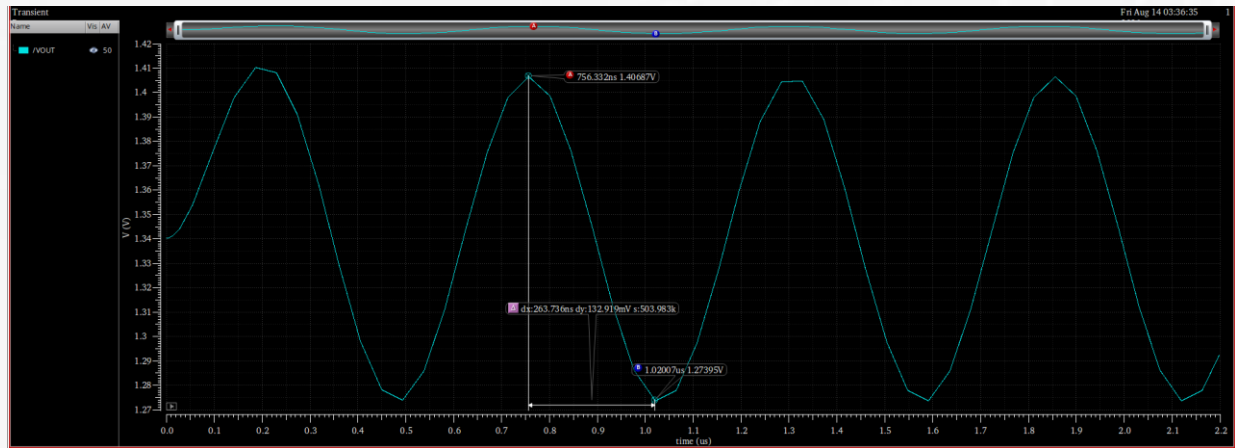


V_{id} : P_2P voltage $\approx 4.26mv$

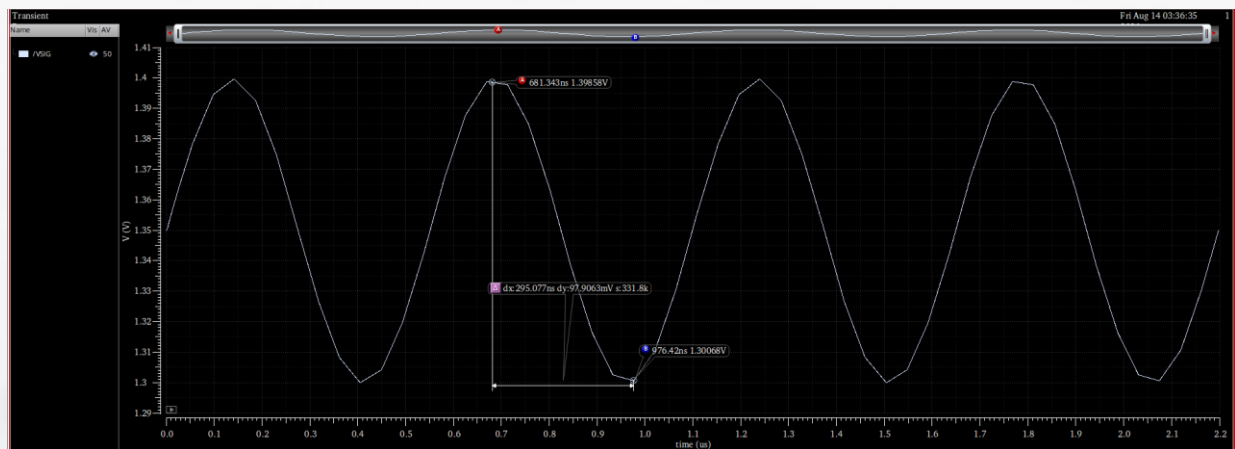
What is the relation between the output and (VP – VN)? Comment

$$\frac{V_{out}}{V_{id}} = \frac{190mv}{4.26mv} \approx 44.6, \text{ very close to the ideal open loop gain } 50$$

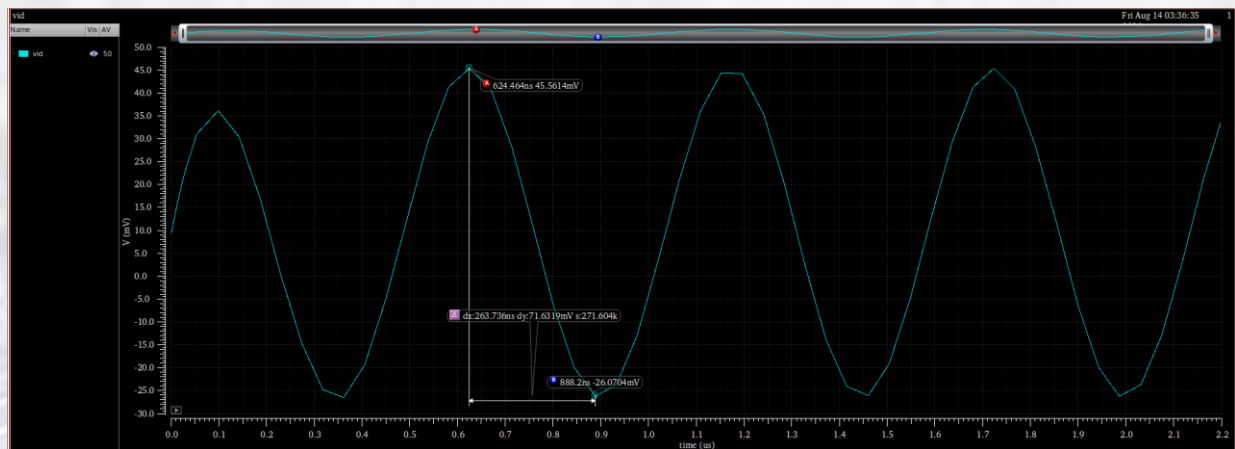
The relation between the output and the differential input is the open loop gain



V_{out} : P_2P voltage $\approx 133\text{mv}$



V_{sig} : P_2P voltage $\approx 98\text{mv}$



V_{id} : P_2P voltage $\approx 71.63\text{mv}$

What is the relation between the output and the input signal?

$$\frac{V_{out}}{V_{sig}} = \frac{133\text{mv}}{98\text{mv}} \approx 1.36 \equiv \text{closed loop gain}$$

What is the relation between the output and (VP – VN)?

$$\frac{V_{out}}{V_{id}} = \frac{133\text{mv}}{71.63\text{mv}} \approx 1.86 \equiv \text{open loop gain, since the open-loop and closed-loop asymptotes coincide at this high frequency.}$$

Compare between this case and the case of 1kHz input.

For 1KHz case, the open loop gain is almost as the ideal gain, but as the frequency increase the open-loop and closed-loop asymptotes coincide at higher frequency, so in case the frequency same as closed loop BW, the open loop gain almost matches the closed loop gain.

